Exploring the Effect of Human Activities on Ocean Temperature, pH, and Turbidity

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***Abstract*— This paper presents the results of an experiment designed to study whether ocean water quality indicators, temperature, pH, and turbidity, vary with distance from shore at Dana Point, California. Because human activity is generally higher near shore, distance from shore was used as a proxy for human impact. This experiment is carried out using a floating autonomous vehicle equipped with a thermistor, a pH sensor, and a turbidity sensor. This vehicle was deployed at Dana Point, California, and traveled from 5 meters from shore to around 35 meters away. During deployment, the retrieved temperature was approximately 18.62°C, which closely matched the 18.7°C ground-truth measurement. The retrieved pH was approximately 7.3–7.5 compared with a ground truth of 8, while the 90° turbidity reading was approximately 100 NTU compared with a ground truth of 13.3 NTU, indicating that the turbidity sensor was not reliable under deployment conditions. Two-sample t-tests comparing the first and last stable measurement windows, corresponding to 5 m and 35 m from shore showed statistically significant differences for temperature, pH, and turbidity. These results suggest that water-quality measurements changed measurably with distance from shore, but because turbidity calibration was inaccurate and distance from shore is only a proxy for human activity, the experiment supports correlation rather than direct causation.**

Keywords—human activity, temperature, pH, turbidity, distance, floating autonomous vehicle

I. Introduction

The goal of this experiment is to study the effect of human activities on temperature, pH, and turbidity of ocean water by collecting and comparing data at different distances from the shore.

This experiment has a wide range of potential applications in marine biology. Coral reefs bleach under warm sea temperatures [1]. Lower pH leads to ocean acidification, which destroy calcified shells in animals such as shellfish. Higher turbidity reduces sunlight penetration which restricts photosynthesis of aquatic plants, leading to a cascading effect that affects the entire marine ecosystem [2]. Thus, studying the effect of human activities on those indicators can inform future policy decisions about how humans are allowed to interact with the ocean to protect it and its inhabitants. Since human activities mainly remain near shore, the experiment will collect data from 5 meters away from shore, where human activities are the most active, to around 35 meters away, where human activities significantly decrease. The data collected at 5 meters and 35 meters will then be processed and analyzed to report any difference.

The rest of the report will present the detailed designs of the robot and the methods and results of the experiment. The Sensor Design section will describe the electrical design of the sensors. The Robot Design section will describe the mechanical design of the robot, including sensor placements. The Deployment Details section will present the methods used to carry out the experiment and measurements. And lastly, the Results section and the Conclusion section will analyze the data collected and discuss its significance.

II. Sensor Design

The sensors chosen are a thermistor which will measure temperature, a pH sensor to measure pH, and a turbidity sensor made out of two photodiodes and an IR light to measure turbidity. The temperature sensor is chosen between a linear temperature sensor, a thermistor, and a thermocouple. The thermistor is chosen based on settling time and implementation complexity considerations. For all sensors, a sampling rate of 10 Hz is used, which is sufficient for the sensors to function properly and sufficient data to be collected. Below are each sensor implementation in detail.

A. Thermistor

The thermistor is a Murata NXFT15WB473FA2B150 [3]. Below is the electrical interface defined, with V_{out} going into the Teensy.

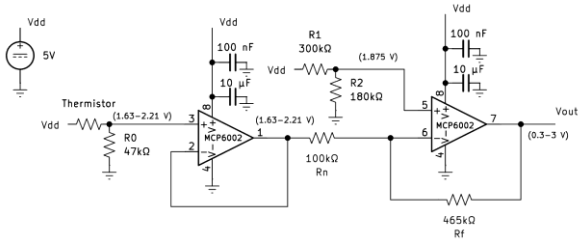


Fig. 1. Temperature sensor schematic.

R_0 is selected to be 47k since this is the nominal thermistor resistance. This gives

$$V_{out1} = \frac{R_0}{R_0 + R_t} V_{dd}, \tag{1}$$

where R_t is the resistance of the thermistor, which varies with the temperature, characterized by this equation:

$$R_t = R_0 * exp\ (B * \left(\frac{1}{T} - \frac{1}{T_0}\right)), \tag{2}$$

where $B = 4050$ K and $T_0 = 298.15$ K. The ocean water temperature at Dana Point is expected to be between 10°C and 20°C. Plugging in the temperature values into Eq. 2, and substituting the calculated R_t values into Eq. 1, V_{out1} is calculated to be between 1.1194 V and 2.2117 V. This expected output range is scaled and offset to the 0.3-3 V range for the Teensy inputs, maximizing the voltage range while leaving a 0.3 V buffer to avoid clipping and non-idealities. An offset amplifier is used to achieve this. In Fig. 1, the formula for the offset amplifier is

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$$V_{out} = (1 + \frac{R_f}{R_{n1}})V_+ - \frac{R_f}{R_{n1}}V_-, \quad (3)$$

with a gain $G = -\frac{R_f}{R_{n1}}$. After setting up a system of equations and solving for gain and V_+, $G = -2.47 = -\frac{R_f}{R_{n1}}$, and $V_+ = 1.66\ V$. Thus, to satisfy those two constraints, $R_f = 247\ k\Omega$, $R_{n1} = 100\ k\Omega$, as well as $R_1 = 20\ k\Omega$, $R_2 = 10\ k\Omega$ which forms a voltage divider that divides V_{dd} supply $V_+ = 1.66\ V$.

In summary, the expected voltage input into the offset amplifier is between 1.1194 V and 2.2117 V, and the electrical interface designed produces an output of 0.3-3 V.

B. pH Sensor

The pH sensor used is a HNNUY Liquid PH 0-14 Value Sensor model Thincol1e9pgoihuw [4], represented as "pH Electrode" in Fig. 2. Fig. 2 contains the pH electrical interface designed, with V_{out} going into the Teensy

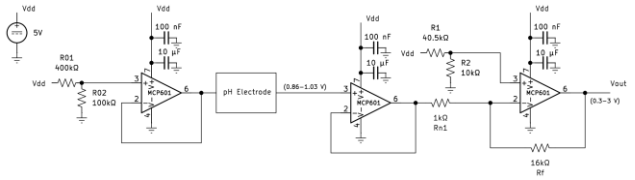


Fig. 2. pH sensor schematic.

$R_{01} = 400\ k\Omega$ and $R_{02} = 100\ k\Omega$ to form a voltage divider to supply 1 V at port 3 of the first op-amp. The first op-amp forms a unity gain buffer, supplying 1 V to the reference pH electrode to offset the pH electrode reading by 1 V. The pH reading is dependent on the temperature. The temperature range at Dana point is estimated to be around 12-18°C, and the pH range is estimated to be around 6.5 to 9.5. An equation relating temperature, pH, and the voltage output from the pH sensor is provided by a Texas Instrument AN-1852 design report [5]:

$$\text{Voltage} = 1 - \frac{8.31451 \times T \times ln(10) \times (pH - 7)}{96485.309}, \quad (4)$$

where T is the temperature of the solution and pH is the pH of the solution. Given the expected ranges of temperature and pH, the expected voltage output from the pH sensor after the offset will be 0.8585-1.0289 V. The lower bound corresponds to the sensor being dipped in a pH 9.5 solution at 18°C and the upper bound corresponds to the sensor being dipped in a pH 6.5 solution at 18°C. This expected output range is then scaled to the 0.3-3 V range for the Teensy inputs, maximizing the Teensy input voltage range while leaving a 0.3 V buffer to avoid clipping and non-idealities. An offset amplifier is used to achieve this. In Fig. 2, the formula for the offset amplifier is

$$V_{out} = (1 + \frac{R_f}{R_{n1}})(\frac{R2}{R2+R1})V_{dd} - \frac{R_f}{R_{n1}}V_-. \quad (5)$$

The gain is thus $G = -\frac{R_f}{R_{n1}}$. After setting up and solving a system of equations, it is obtained that $G = 15.9$, and the resistor values that satisfy Eq. 5 are $R_f = 16\ k\Omega$, $R_{n1} = 1\ k\Omega$, $R_1 = 40.8\ k\Omega$, $R_2 = 10\ k\Omega$. However, since a resistor value of $40.8\ k\Omega$ was not able to be achieved given the resistors available at time of circuit construction, R_1 is actually constructed to be $40.5\ k\Omega$, as shown in Fig. 2.

In summary, the expected voltage input into the offset amplifier from the pH electrode is between 0.8585 V and 1.0289 V, and the electrical interface designed produces an output of 0.3-3 V.

C. Turbidity Sensor

The turbidity sensor is made from an Infrared diode that emits infrared light periodically, and two photodiodes that detects infrared light, with one placed directly across the IR diode and the other 90° from the IR diode. The two photodiodes will be referred to hereafter as the 180° turbidity sensor the 90° turbidity sensor, respectively. To emit light periodically, the IR diode needs to be powered by a square wave. Thus, the first part of this schematic will be a square wave generator using a 555 timer, as shown in Figure 3.

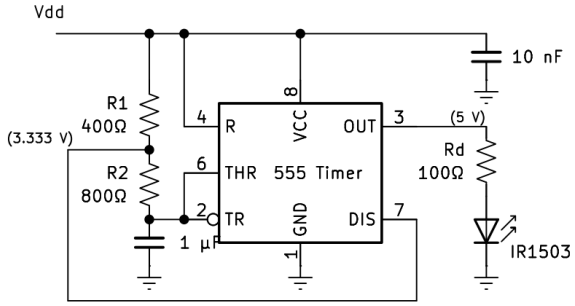


Fig. 3. 555 timer square wave generator schematic.

This circuit creates a periodic signal (square wave) with a frequency of 720 Hz and a duty cycle of 60% using the 555 timer and analog elements, such as resistors and capacitors.

The formula that describes the frequency of the oscillations is

$$f = \frac{1.44}{(R_1 + 2R_2)C_1}. \quad (6)$$

The formula that describes the duty cycle of the oscillator is

$$duty\ cycle = \frac{R_1 + R_2}{R_1 + 2R_2}. \quad (7)$$

Choosing C to be 1µF, R_1 and R_2 are solved in a system of equations (Eq. 6 and Eq. 7) by plugging in f = 720 Hz and duty cycle = 0.6. The solution has $R_1 = 400\ \Omega$ and $R_2 = 800\ \Omega$.

The formula that describes the current through the IR diode is

$$I = \frac{V_{out} - 1.2V}{R_d}, \quad (8)$$

where the 1.2 V in Eq. 8 is the voltage drop across the IR diode. The current through the IR diode needs to be between 20 and 40 mA. We decided on I = 38 mA. Knowing $V_{out} = V_{dd}$ = 5V, we used Eq. 8 to calculate that R_d = 100 Ω.

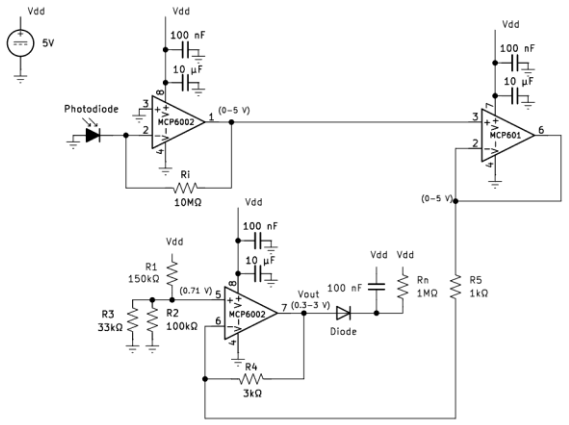


Fig. 4. 90° turbidity sensor schematic.

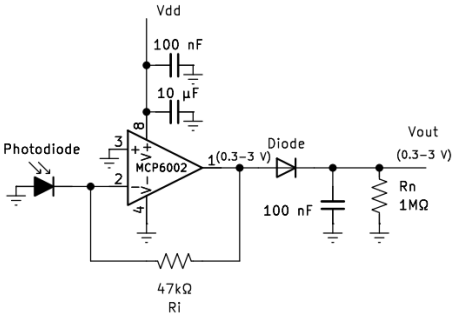


Fig. 5. 180° turbidity sensor schematic.

In both photodiode circuits, the initial design for the resistor values did not behave as expected. The values were changed experimentally with assistance from Xavier Walter to achieve the designs shown in Fig. 4 and Fig. 5.

Experimentally, the voltage output of the transimpedance amplifier which converts the current generated by the photodiode into a voltage are 0-5 V and 0.3-3 V, for the 90° turbidity sensor and the 180° turbidity sensor, respectively. Since the 90° transimpedance output is not within the Teensy range, it is scaled and offset into the 0.3-3 V range for the Teensy inputs, maximizing the Teensy input voltage range while leaving a 0.3 V buffer to avoid clipping and non-idealities. Moreover, a min-detector is added to the 90° turbidity sensor circuit, and a max-detector is added to the 180° turbidity sensor circuit to produce a DC voltage output instead of a square wave. The 90° turbidity sensor uses a min-detector instead of a max-detector because the offset amplifier in the circuit is inverting, whereas the 180° turbidity circuit only has a transimpedance amplifier, which is non-inverting.

In summary, the voltage input from the 90° turbidity sensor is 0-5 V and the voltage input from the 180° turbidity sensor is 0.3-3 V, and their circuits designed both produce an output of 0.3-3 V.

III. ROBOT DESIGN

Below is a diagram of the mechanical design of the robot and sensor integration.

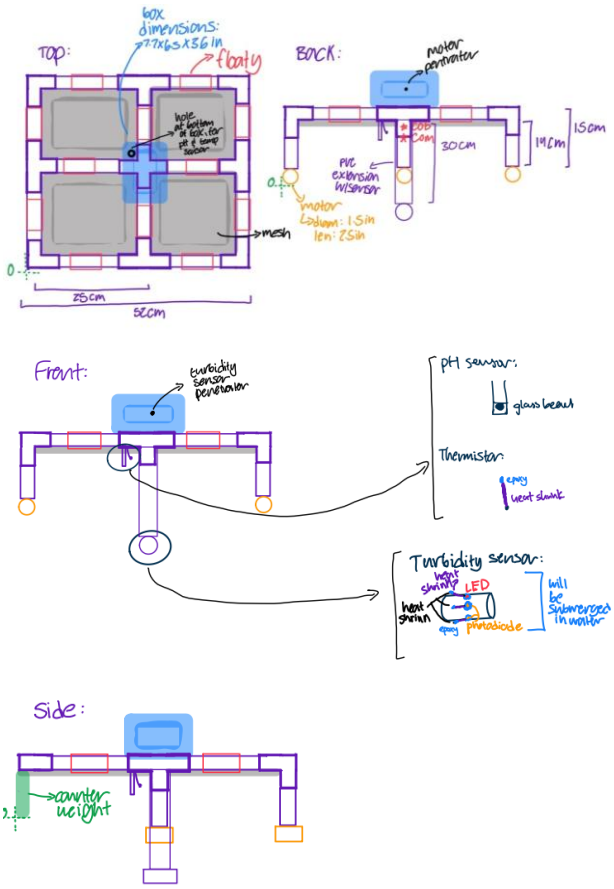


Fig. 6. Mechanical drawing of the robot (top, back, front, and side view).

The mass of the robot is calculated by weighing each component individually and summing the masses up. The total robot weight ended up being 1789.24 grams.

The center of mass and center of buoyancy were calculated through the use of the following formulas:

$$Com = \sum(w_i * d_i)/\sum w_i \quad (9)$$

$$Cob = \sum(v_i * d_i)/\sum v_i \quad (10)$$

Where w_i is the weight of each component and d_i is distance away the center of mass of each component from the defined zero (9). The center of mass along the axis of symmetry in the top view is calculated to be 26 cm, and the side view, is calculated to be 14.09 cm. Where v_i is the volume of each component and di is distance away the center of mass of each component from the defined zero (10). Again, the center of buoyancy along the axis of symmetry in the top view is

calculated to be 26 cm, and the side view is calculated to be 14.94 cm. It is important to note that in the z-direction the center of buoyancy is above the center of mass and they are both along the same axis. This indicates that the robot will be stable once in water.

Since the floating autonomous vehicle is a surface robot, it does not need to be neutrally buoyant, thus no ballasting calculations was necessary. Instead, floaties were placed symmetrically onto the robot frame to ensure high buoyancy and prevent yaw and roll. The robot has a low profile against the wind through its flat design, which minimizes the above-water surface area. The robot also has have a low profile against currents since its base has a larger surface area which will prevent the current from moving it in undesirable ways.

Confirming using the Pythagorean theorem, the distance between the motors and the IMU satisfies the > 30 cm constraint. Additionally, as shown in Fig. 6, the turbidity sensor is centered and placed 14 cm below the frame where the box is to ensure it is fully submerged in the water and to minimize surface lighting interference. The pH sensor and thermistor are directly below the IMU box, both cut through the bottom of the box and waterproofed and secured with marine epoxy.

Everything inside of the IMU box are waterproofed. Each penetrator is also epoxied to ensure no leakage occurs. For the pH sensor, the end with a cylinder of a wider radius will be placed in the water, which is already waterproof. Any exposed soldered joints will be waterproofed with heat shrink. As for the turbidity sensor, a special casing made out of PVC where three holes are drilled into for the LED and photodiodes as seen in Fig. 6. The brim of these holes are epoxied, and the exposed wires are heat shrunk. Any exposed soldered joints are waterproofed with heat shrink. The bead of the thermistor is placed into the water, and the copper colored wire is waterproofed with heat shrink. Again, any exposed soldered joints are waterproofed with heat shrink.

The robot is a surface robot, and is designed to go straight in one direction only. Thus, all three motors face the same direction. The robot is first test-deployed at pHake lake, and then officially deployed at Dana point. Between those two deployments, four more floaties are added to increase buoyancy.

IV. DEPLOYMENT DETAILS

A calibration curve was created for each sensor after the robot was constructed to relate the voltage reading to the actual temperature, pH, and turbidity readings.

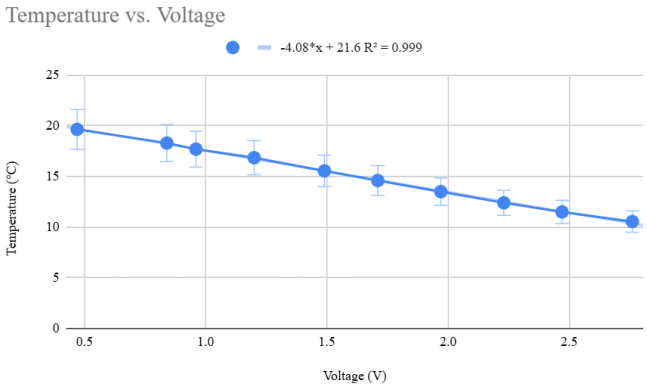


Fig. 7. Temperature sensor calibration curve.

For temperature, the thermistor was dipped into water ranging from 10-20°C (ground truth temperature based on the average of two digital thermometers) and the voltage output was read from the serial monitor. The data fit a linear calibration curve with $R^2 = 0.999$, indicating a strong linear relationship, as shown in Fig. 7. The equation of the linear calibration curve relating voltage output to temperature is:

$$Temperature\ (°C) = -4.08 \times Voltage + 21.6. \quad (11)$$

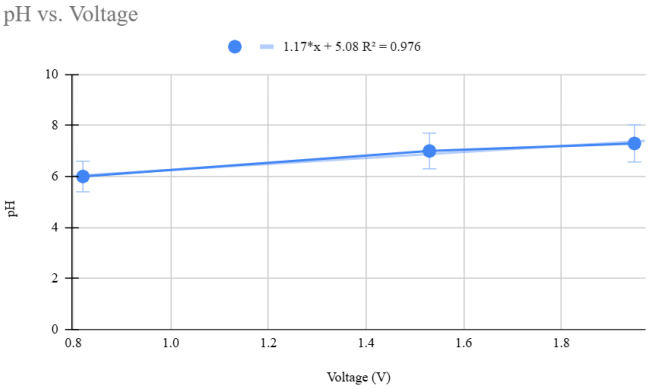


Fig. 8. pH sensor calibration curve.

For pH, solutions of pH 6.5, 7, and 9 were created using solutions with a known pH. The probe was dipped into these solutions and the voltage output from Teensy on the Arduino serial monitor was recorded. A linear calibration curve was fit to the points with $R^2 = 0.976$, indicating a strong linear relationship, as shown in Fig. 8. The equation of the linear calibration curve relating voltage output to pH is:

$$pH = 1.17 \times Voltage + 5.08. \quad (12)$$

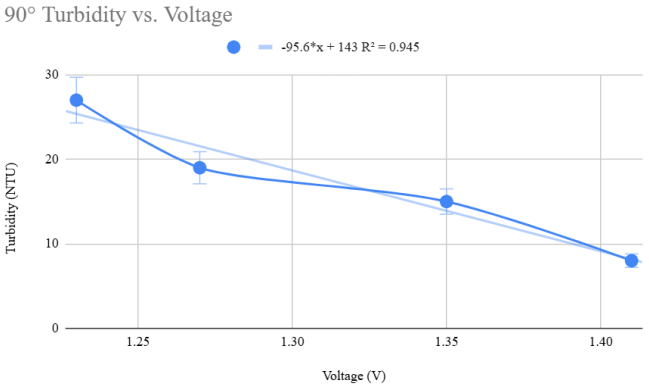


Fig. 9. 90° turbidity sensor calibration curve.

For turbidity, the sensor was moved from a horizontal to vertical position to fit inside the cups of different turbidity solutions (8 NTU, 16 NTU, 19 NTU, and 26 NTU). In one trial, one side of the casing was tapped and the other trial was in complete light, however neither were representative of ocean conditions. Nevertheless, a linear calibration curve was fitted for the 90° turbidity sensor with $R^2 = 0.945$, indicating a strong linear relationship, as shown in Fig. 9. The equation of the linear calibration curve relating voltage output and turbidity is:

$$Turbidity\ (NTU) = -95.6 \times Voltage + 143. \quad (13)$$

The 180° turbidity sensor, on the other hand, was malfunctioning. It consistently produces 0.2 V across all turbidity solutions, and produces approximately 1.16 V when it is not placed in a solution. The malfunctioning could be due to a waterproofing issue, since the sensor does respond to being placed in water, but produces the same voltage across all turbidities.

These calibration curves are used to 1. Verify that the sensors are working properly and 2. Relate the voltage readings from data collected to their corresponding physical property values.

The deployment process is as follows: the robot is first placed in water, then the box is opened to turn the motherboard, then the box is closed and sealed during the 10 second period the motor is off initially. The team kayaked alongside the robot during its trajectory. At the end of deployment, the team fishes the robot out of the water, turns the mother board off, and then retrieves the SD card to read the data. The robot was deployed starting 5 m away from shore, and stopped at 35 m away from shore. The motor stops every 5 meters for data collection.

As per deployment safety, all team members signed waivers, were wearing life jackets during kayaking and following all safety precautions. The robot has a tether tied to it to allow the team members retrieve the robot safely and/or redirect it when an obstacle is close by. Moreover, the robot is only handled in the kayak by dry hands, minimizing risk of electric shock or shorting the motherboard.

All phones were left on land during Dana Point deployment, so no pictures of the robot or robot trajectory was captured at Dana Point. However, a photo of the robot during testing at pHake lake and a general photo of the beach at Dana Point were captured, both shown below in Fig. 10.



Fig. 10. (a) On the left is a photo of the robot deployed at pHake lake. (b) On the right is a deployment location at Dana Point.

V. Results

Below are the data collected from each sensor at Dana Point:

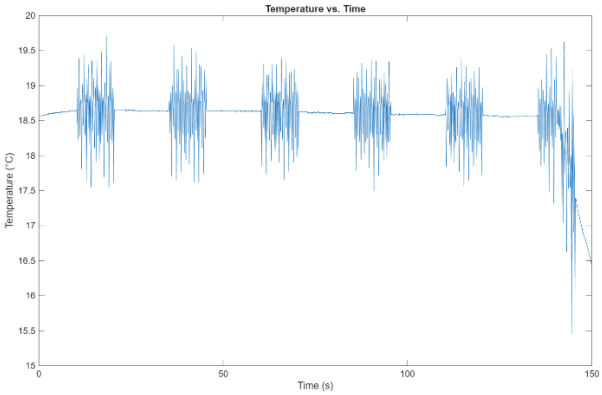


Fig. 11. Temperature data.

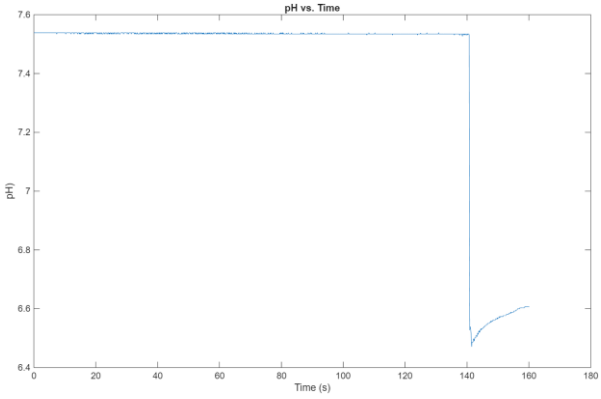


Fig. 12. pH data.

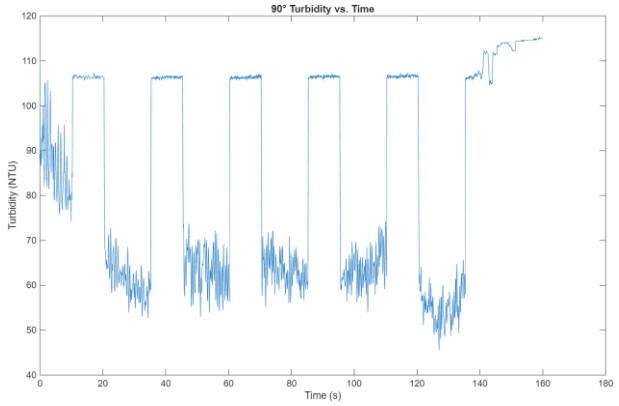


Fig. 13. 90° turbidity sensor data.

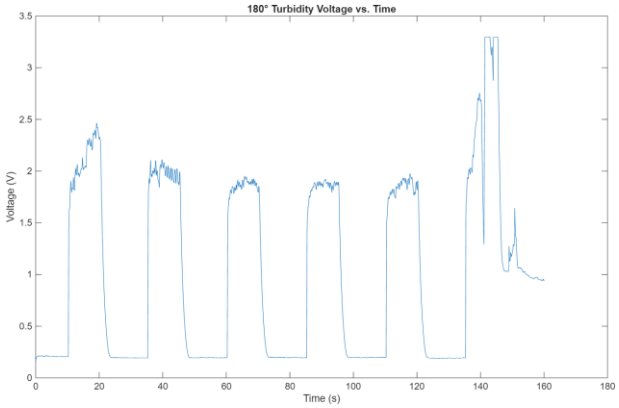


Fig. 14. 180° turbidity sensor data. The data are shown in voltage rather than NTU because the sensor did not produce a valid calibration curve.

The periodic noise present in all graphs result from operating motors. The stable parts are when the motors stops and the robot remains in place to collect data. Thus, the data of importance are the stable parts.

Based on the calibration curves and looking at the stable parts, the temperature of the ocean is approximately 18.62°C, the pH of the sea water is between 7.3 and 7.5, and the turbidity of the water is approximately 100 NTU. The ground truth measurement for temperature, pH, and turbidity are 18.7°C, 8, and 13.3 NTU. The temperature measurement closely matched the ground truth, the pH measurement was less accurate, and the turbidity measurement was inaccurate. One possible explanation for the inaccuracy of the 90° turbidity sensor is that its calibration environment did not fully resemble the sea water condition at Dana Point, as mentioned before in Deployment Details section. And as mentioned in the same section, the 180°C turbidity sensor likely had a waterproofing issue, causing its malfunction.

Since the goal of this deployment is to study whether those three metrics are different close to shore versus far from shore, The first and last set of stable measurements for each sensor, which corresponds to 5 meters and 35 meters from shore, are compared. A two-sample t-test is conducted for each pair in MATLAB and the results are as follows:

H_0: The measurements made (temperature, pH, or turbidity) at 5 m and 35 m from shore are not significantly different.

H_a: The measurements made (temperature, pH, or turbidity) at 5 m and 35 m from shore are significantly different.

1. Temperature: For a confidence level of $\alpha = 0.05$, a p-value of 8.734×10^{-12} is obtained. Since the p-value is less than α, the null hypothesis is rejected.
2. pH: For a confidence level of $\alpha = 0.05$, a p-value of 1.072×10^{-41} is obtained. Since the p-value is less than α, the null hypothesis is rejected.
3. Turbidity: For a confidence level of $\alpha = 0.05$, a p-value of 1.466×10^{-6} is obtained. Since the p-value is less than α, the null hypothesis is rejected.

Therefore, there is significant evidence to prove that the temperature, pH, and 90° turbidity data collected at Dana Point is different close to and far from shore, showing a correlation between the seawater quality indicators and distance from shore. However, since distance from shore is only a proxy for human activity, this test alone does not directly prove that human activities impact seawater quality.

VI. Conclusion

The data collected at Dana Point provide statistically significant evidence that temperature, pH, and the 90° turbidity sensor readings changed between 5 m and 35 m from shore. The two-sample t-tests produced p-values of 8.734×10^{-12}, 1.072×10^{-41}, and 1.466×10^{-6} for temperature, pH, and turbidity, respectively, so the null hypothesis was rejected for all three measurements. This means that the robot was able to detect measurable changes in water-quality indicators over the deployment path.

However, the results should be interpreted carefully. The temperature sensor was the most reliable sensor because its retrieved value of 18.62°C closely matched the ground-truth value of 18.7°C. The pH sensor was less accurate, measuring approximately 7.3–7.5 compared with a ground truth of 8.0. The 90° turbidity sensor was not accurate, measuring approximately 100 NTU compared with a ground-truth value of 13.3 NTU. This large error likely came from the calibration environment not matching the lighting, sensor orientation, and water conditions at Dana Point. The 180° turbidity sensor also malfunctioned, likely due to a waterproofing issue, so its data could not be converted into meaningful turbidity values. Even though the data collected support the idea that water-quality indicators vary with distance from shore, which is consistent with the hypothesis that near-shore human activity may affect seawater quality, since distance from shore is only a proxy for human activity, this experiment shows correlation but cannot prove on its own that human activity caused the observed changes.

Future versions of this project should focus on improving both sensor reliability and experimental control. First, the turbidity sensor should be recalibrated in conditions that better match ocean deployment, including similar ambient light, sensor orientation, and water movement. Second, the 180° turbidity sensor should be redesigned with stronger

waterproofing and tested for leakage before deployment. Third, the pH sensor should be redesigned or recalibrated over a narrower expected seawater range, such as pH 7–8.5, to improve resolution. Fourth, the robot should record GPS data so that each measurement can be tied to a precise location rather than an estimated distance from shore. Finally, future trials should repeat the deployment at different times of day and across multiple days so that short-term variation can be separated from consistent distance-from-shore trends.

In terms of skills and lessons learned, debugging skills are practiced and improved through this project. The experiment process also shows how calibration and waterproofing are just as important as the sensor circuit design itself.

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